CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Instruction and Addressing Modes

Lecture No.- 04



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Recap of Previous Lecture



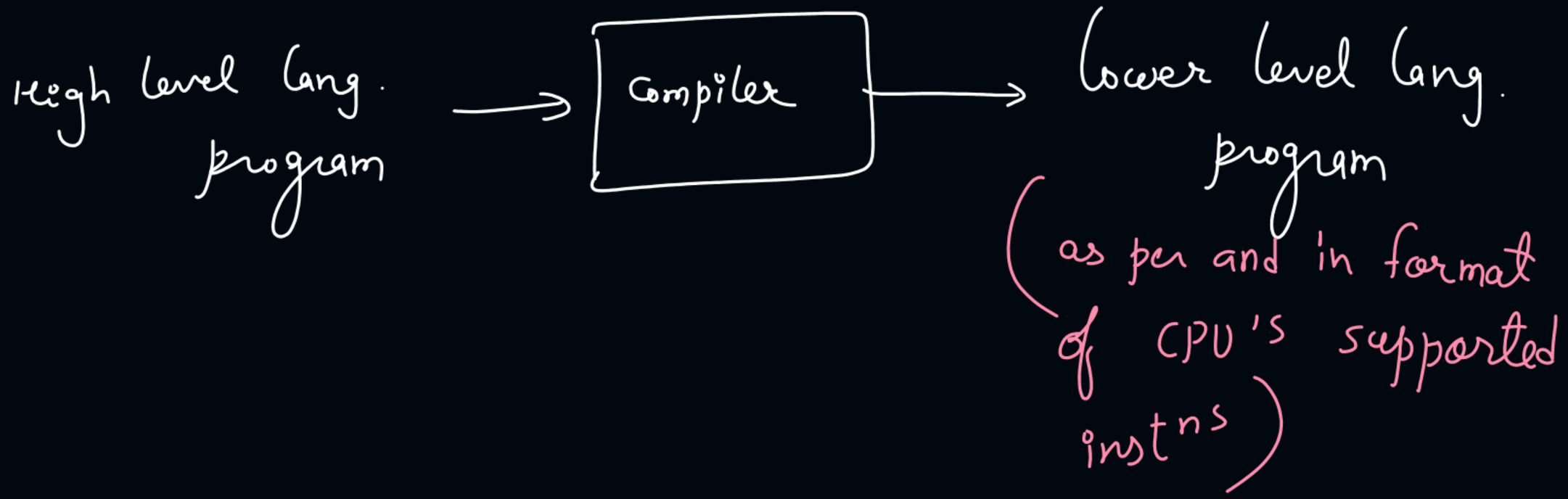
Topic Instruction

Topic Multiple Instruction Support

Topics to be Covered



Topic	Instruction


High level lang. program
Compiler
Lower level lang. program
(as per and in format of CPU's supported instns)

example:-

2-address inst^ns supported in CPU

Reg-based architecture

Programming statement:-

$a = b + c$

a, b, c ⇒ mem. operands

⇒

Instructions

mov R1, b	R1 ← b
mov R2, C	R2 ← C
ADD R1, R2	R1 ← R1 + R2
mov a, R1	a ← R1

mov R1, b → load

mov a, R1 → store

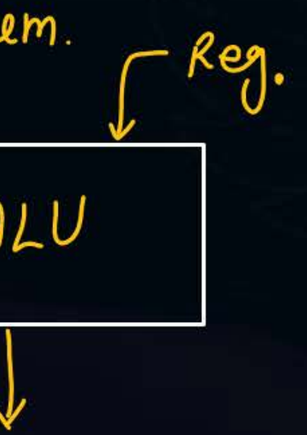
[NAT]

#Q. In a simplified computer the instructions are:

OP R_i, R_j	- Performs R_i Op R_j and stores the result in R_j
OP m, R_i	- Performs val Op R_i and stores the result in R_i val denotes the content of memory location m
MOV m, R_i	- Moves the content of memory location m to register R_i
MOV R_i, m	- Moves the content of register R_i to memory location m

The computer has only two registers and OP is either ADD or SUB. Consider the following basic block:

t1 = a + b
t2 = c + d
t3 = e – t2
t4 = t1 – t3



[NAT]

Ans = 3

#Q. Assume that all operands are initially in memory. The final value of the computation should be in memory. What is the minimum number of MOV instructions in the code generated for this basic block?

R1
$t1 = a + b$
R2
$t2 = c + d$
R2
$t3 = e - t2$
R1
$t4 = t1 - t3$

```
mov  b, R1      R1 ← b
ADD  a, R1      R1 ← a + R1
mov  d, R2      R2 ← d
ADD  c, R2      R2 ← c + R2
SUB  e, R2      R2 ← e − R2
SUB  R1, R2     R2 ← R1 − R2
mov  R2, m      m ← R2
```

[NAT]

#Q. In a simplified computer the instructions are:

OP R_i, R_j	- Performs R_i Op R_j and stores the result in R_i
OP R_i, m	- Performs R_i Op val and stores the result in R_i val denotes the content of memory location m
MOV m , R_i	- Moves the content of memory location m to register R_i
MOV R_i , m	- Moves the content of register R_i to memory location m

The computer has only two registers and OP is either ADD or SUB. Consider the following basic block:

R1 t1 = a + b
R2 t2 = c + d
R2 t3 = e – t2
t4 = t1 – t3

[NAT]

Ans = 5

#Q. Assume that all operands are initially in memory. The final value of the computation should be in memory. What is the minimum number of MOV instructions in the code generated for this basic block?

mov a, R1	R1 ← a
ADD R1, b	R1 ← R1 + b
mov C, R2	R2 ← C
ADD R2, d	R2 ← R2 + d
mov R2, x	x ← R2
mov e, R2	R2 ← e
SUB R2, x	R2 ← R2 − x

SUB R1, R2	R1 ← R1 − R2
mov R1, m	m ← R1

[NAT] GATE-2020

H. W.

#Q. A processor has 64 registers and uses 16-bit instruction format. It has two types of instructions: I-type and R-type. Each I-type instruction contains an opcode, a register name, and a 4-bit immediate value. Each R-type instruction contains an opcode and two register names. If there are 8 distinct I-type opcodes, then the maximum number of distinct R-type opcodes is _____?

[NAT] GATE-2018

H. W.

#Q. A processor has 16 integer registers (R0, R1, … , R15) and 64 floating point registers (F0, F1, … , F63). It uses a 2-byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3, and Type 4. Type-1 category consists of four instructions, each with 3 integer register operands (3Rs). Type-2 category consists of eight instructions, each with 2 floating point register operands (2Fs). Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand (1R+1F). Type-4 category consists of N instructions, each with a floating-point register operand (1F).

The maximum value of N is _____ ?



2 mins Summary

Topic Instructions

Topic Multiple Instruction Support



Happy Learning

THANK - YOU